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Price discovery in the markets for credit risk: a Markov switching approach

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Abstract: We examine price discovery in the Credit Default Swap and corporate bond market. Using a Markov switching framework enables us to analyze the dynamic behavior of the information shares during tranquil and crisis periods. The results show that price discovery takes place mostly on the CDS market. The importance of the CDS market even increases during the more volatile crisis periods. According to a cross sectional analysis liquidity is the main determinant of a market's contribution to price discovery. During the crisis period, however, we also find a positive link between leverage and CDS market information shares. Overall the results indicate that price discovery measures and their determinants change during tranquil and crisis periods, which emphasizes the importance of more flexible frameworks, such as Markov switching models.

Keywords: corporate bond; credit default swap; Markov switching; price discovery; unique information share.

JEL classification: C14; G15.

1 Introduction

The markets for credit risk have recently attracted great attention in finance literature. Particularly, the reaction of the credit default swap (CDS) market to the crisis period of 2007–2009 has been analyzed, as it allows researchers to derive information that may help to maintain financial market stability in more volatile periods. In this context, information transmission between markets is of paramount interest. One of the key questions is how the price discovery process reacts to changes in the market environment, such as a serious financial crisis. In order to examine this issue, we use a Markov switching framework to analyze the dynamics of the price discovery process between the markets for credit risk, i.e. the CDS and corporate bond markets, during tranquil and crisis periods. Furthermore, we examine potential factors that influence a market's importance for the price discovery process and whether these change during more volatile times.

The pronounced regime-specific behavior displayed by CDS spreads has previously been documented by Alexander and Kaeck (2008) and Leppin and Reitz (2014), among others. Both studies report changes in the dynamics of CDS prices depending on high- and low-volatility regimes. However, previous price discovery studies have neglected this characteristic. Blanco Brennan, and Marsh (2005), Dimpfl and Peter (2013), and Grammig and Peter (2013) examine the contribution of the CDS and corporate bond market to the price discovery process. All of them determine the CDS market as the leading one. While Dimpfl and Peter (2013) use transfer entropy, Blanco, Brennan, and Marsh (2005) apply the common measure of Hasbrouck (1995) information shares to quantify a market's contribution to the common efficient price. Hasbrouck's information share measure was modified by Grammig and Peter (2013) to solve the problem of indeterminacy inherent in the variance decomposition approach and to ultimately propose a unique information share. Our approach extends their methodology by using a Markov switching framework that models CDS prices

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and bond spreads depending on two different variance regimes. This approach renders a unique information share measure along the lines of Hasbrouck (1995) and at the same time accounts for regime-dependent behavior. We thereby rely on a model recently proposed by Herwartz and Lütkepohl (2014), which identifies shocks in a autoregressive system with Markov switching by combining conventional with statistical identification methods. Applying their model to the context of price discovery allows a much more accurate assessment concerning the informationally leading market.

We analyze CDS and bond spread time series on 22 European iTraxx reference entities during the period from January 2004 to June 2015. Consistent with previous studies we find that the CDS market leads price discovery for most reference entities. During more volatile periods the leadership of the CDS market becomes even more pronounced. As regards potential determinants of information shares we find that relative liquidity matters. Other factors like leverage, market capitalization, or industry affiliation do not seem to explain the information share.

The remainder of the paper is organized as follows. Section 2 briefly outlines the Hasbrouck (1995) information shares and introduces the Markov switching model and the corresponding information shares. Section 3 describes the data and estimation procedure. In Section 4 we present the estimated information shares and their determinants derived from a cross-sectional regression. Section 5 concludes.

2 A Markov switching approach to estimate unique information shares

In the following we briefly outline the Hasbrouck (1995) methodology and subsequently introduce the Markov switching model. Assuming pairwise cointegrated prices in n markets, price dynamics can be described by a Vector Error Correction Model (VECM; see Hasbrouck 1995):

$$\Delta p_t = \alpha\beta' p_{t-1} + \Gamma_1 \Delta p_{t-1} + \ldots + \Gamma_{q-1}\Delta p_{t-q+1} + u_t, \tag{1}$$

where β is the $n\times n$ cointegration matrix, α denotes an $n\times n$ matrix that contains the adjustment coefficients, and the Γs are $n\times n$ parameter matrices that govern the autoregressive behavior of the price series. Furthermore u_t are mean-zero composite innovations with covariance matrix Σ_u. Within the Hasbrouck (1995) framework, the composite innovations are modeled as a linear combination of *iid* idiosyncratic innovations, i.e $u_t = B\varepsilon_t$, where ε_t is a mean-zero random variable with covariance matrix I_n. B denotes an $n\times n$ matrix whose elements capture the contemporaneous effect of the idiosyncratic innovations.

According to Hasbrouck (1995) the common efficient price p_t^* follows a random walk, where v_t are the efficient price innovations. Assuming that initial prices are zero $(p_0^*=0)$, the efficient price series is related to the VECM parameters via the infinite sum of innovations as

$$p_t^* = v_t + v_{t-1} + \ldots = \xi' B(\varepsilon_t + \varepsilon_{t-1} + \ldots).$$

ξ constitutes the vector of long-run impact coefficients of the idiosyncratic innovations, which can be derived as the common row vector of $\Xi = \beta_\perp \left[\alpha_\perp' (I_n - \sum_{i=1}^{q-1}\Gamma_i)\beta_\perp\right]^{-1}\alpha_\perp'$, where $\alpha_\perp$ denotes the orthogonal complement of α (see Johansen 1995).

Hasbrouck's information shares are then defined as the share in the variance of the efficient price innovation that is attributed to each of the markets. Consequently, decomposing the efficient price innovation variance given by

$$\mathrm{Var}(v_t) = \xi' \Sigma_u \xi = \xi' BB'\xi$$

results in the vector of information shares (IS)

$$IS = \frac{[\xi' B]^{\circ 2}}{\xi' BB'\xi}, \tag{2}$$

where $[\cdot]^{\circ 2}$ denotes element-wise squaring (Hadamard power) of the vector given in brackets.

The main drawback of this approach is that the matrix B, which contains the contemporaneous effects, cannot be identified. From Equation (1) we can estimate the covariance matrix of the composite shocks Σ_u, which is related to B by $\Sigma_u=BI_nB'$. However, as Σ_u is symmetric, we can only identify $(n^2+n)/2$ of the n^2 elements in B.

To overcome this issue, Hasbrouck (1995) proposes to use a Cholesky decomposition of Σ_u. Thereby we replace the matrix B with a lower triangular matrix C such that $\Sigma_u=CI_nC'$. The Hasbrouck information shares are then given by

$$HIS=\frac{[\xi' C]^{\circ 2}}{\xi' CC'\xi}. \tag{3}$$

Applying the Cholesky decomposition involves ruling out certain contemporaneous effects. The ordering of the markets becomes important as shocks in the market ordered first can instantly influence all other markets, while the contemporaneous effects of the market ordered last are restricted to zero. This results in an upper bound estimate for the information share of the market ordered first and a lower bound estimate of the market ordered last. Permuting the ordering yields lower and upper bounds for each market.

The distance between the upper and lower bound of the information share can be large depending on the amount of contemporaneous correlation between the composite innovations u_t (compare Booth, Lin, Martikainen, and Tse, 2002; Hupperets and Menkveld 2002). As a result, conclusions regarding the leading market are only vague. Grammig and Peter (2013) resolve this problem by assuming a mixture normal distribution for the composite innovations. This assumption is supported by the commonly observed fat tails of financial return distributions (see Longin and Solnik 2001; Rigobon 2003; Lanne and Lütkepohl 2010), which we also observe in our data. Figure 1 presents the residuals of the VECM in Equation (1) for three reference entities (RWE, Vattenfall, and Volkswagen). The heavy tails are particularly pronounced for the bond equation, but all distributions are more peaked than what would be suggested by a normal distribution. The resulting tail-dependence ultimately allows for the identification of contemporaneous effects and unique information shares. In the context of Grammig and Peter (2013), the distributional assumption implies two regimes that are associated with different variances. The regimes are assumed to be independent.

In the current approach we use a less restrictive assumption concerning the regimes. We assume that the two regimes are not independent, but depend on a Markov process s_t (see Lanne, Lütkepohl, and Maciejowska 2010; Herwartz and Lütkepohl 2014). Furthermore we assume that conditional on state s at time t, the composite innovations u_t are normally distributed with covariance Σ_u^s, i.e. $u_t|s_t\sim\mathcal{N}(0, \Sigma_u^s)$, where s_t is generated by a discrete Markov chain of first order with transition probabilities: $p_{ij}=P(s_t=j|s_{t-1}=i)\forall i, j\in\{1, 2\}$.[1]

The idiosyncratic innovations e_t are either from state 1 or state 2, where the states are described by different variances and therefore

$$e_t=\begin{cases} e_{1,t}\sim N(0, I_n) & \text{with probability } p \\ e_{2,t}\sim N(0, \Psi) & \text{with probability } 1-p, \end{cases} \tag{4}$$

where $p=P(s_t=1)=\frac{1-p_{22}}{2-p_{11}-p_{22}}$ is the unconditional probability of regime 1 (see Hamilton 1994). Ψ is a diagonal matrix with distinct elements. It follows that the composite innovations are given by $u_t=B\varepsilon_t=We_t$ with covariance matrix $\Sigma_u=W\Sigma_eW'=W(pI_n+(1-p)\Psi)W'$. In accordance with Equation (2), the vector of information shares is now derived as

$$IS=\frac{\left[\xi' W(pI_n+(1-p)\Psi)^{0.5}\right]^{\circ 2}}{\xi' W(pI_n+(1-p)\Psi)W'\xi}. \tag{5}$$

1 Modeling regime (or time) dependent VECM parameters is possible (compare Doetz 2007), however, it is not conformable with the microstructure model at the base of the Hasbrouck (1995) methodology. Therefore, the economic meaning of information shares derived from such a framework is questionable.

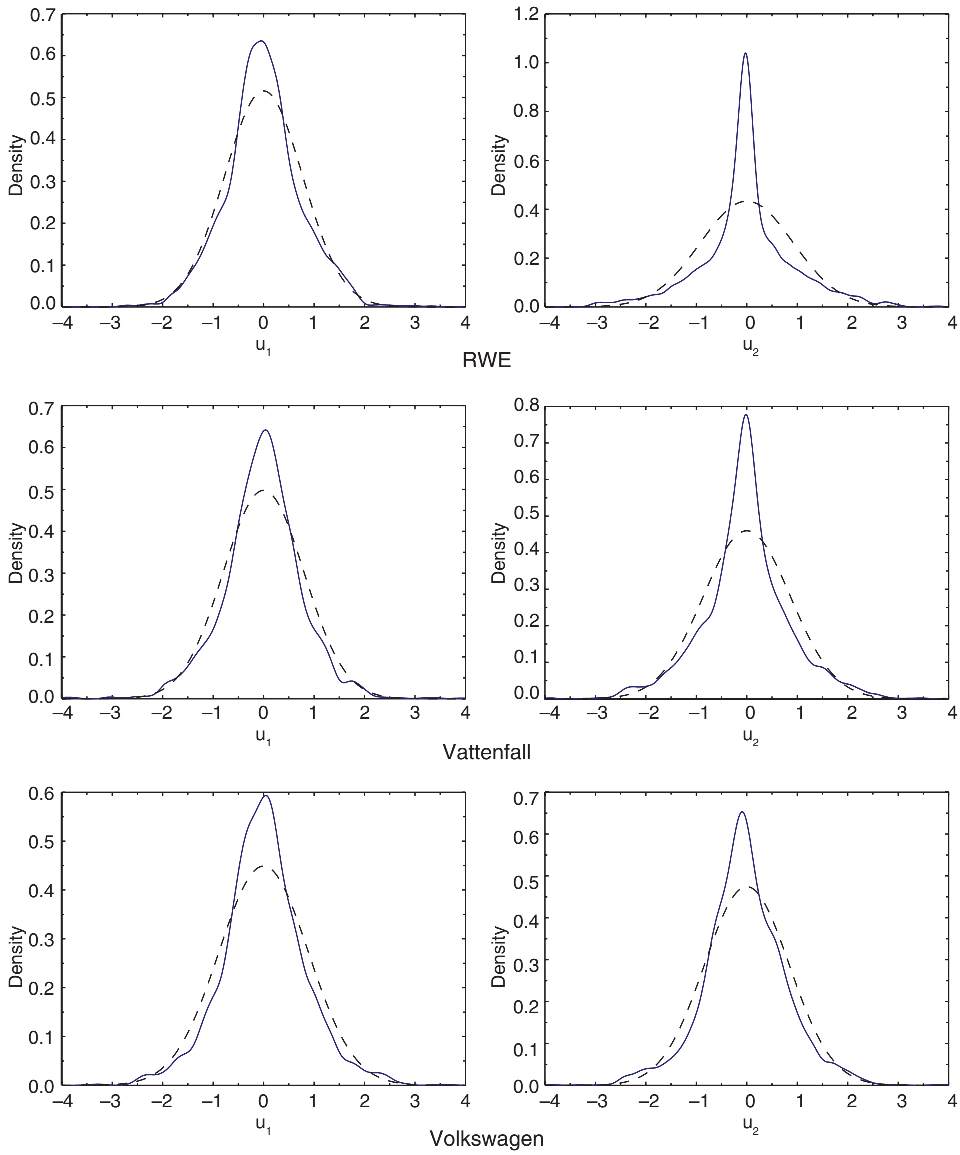

Figure 1: VECM residuals.
The figure presents kernel density plots of the VECM residuals (solid line) for the three entities RWE, Vattenfall, and Volkswagen. The dashed line represents the cdf of a normal distribution with corresponding mean and variance estimated from the residuals.

The identification of the contemporaneous effects within this framework requires certain assumptions regarding the covariance matrix of the second regime Ψ, as well as the matrix of contemporaneous effects W. The variances of the variables in question, i.e. the diagonal elements of Ψ, have to be distinct (for a formal proof see Lanne, Lütkepohl, and Maciejowska 2010). Intuitively, this assumption should be fulfilled: the time series plots in Figure 2 suggest that the bond data are more volatile than the CDS data during some periods of the sample. Furthermore Grammig and Peter (2013) show that for unique identification additional restrictions have to be imposed on W. These restrictions are

$$w_{i,i}>0 \ \forall i \text{ and } w_{i,i}>|w_{j,i}| \ \forall j\neq i,$$

where $w_{i,j}$ denotes the row i, column j element of W. These restrictions arise naturally within the price discovery framework, where we observe n prices that rely on the same underlying asset. They imply that the effect of an idiosyncratic price shock in market i ($e_{i,t}$) on the price observed in this very market ($p_{i,t}$) has to be of the

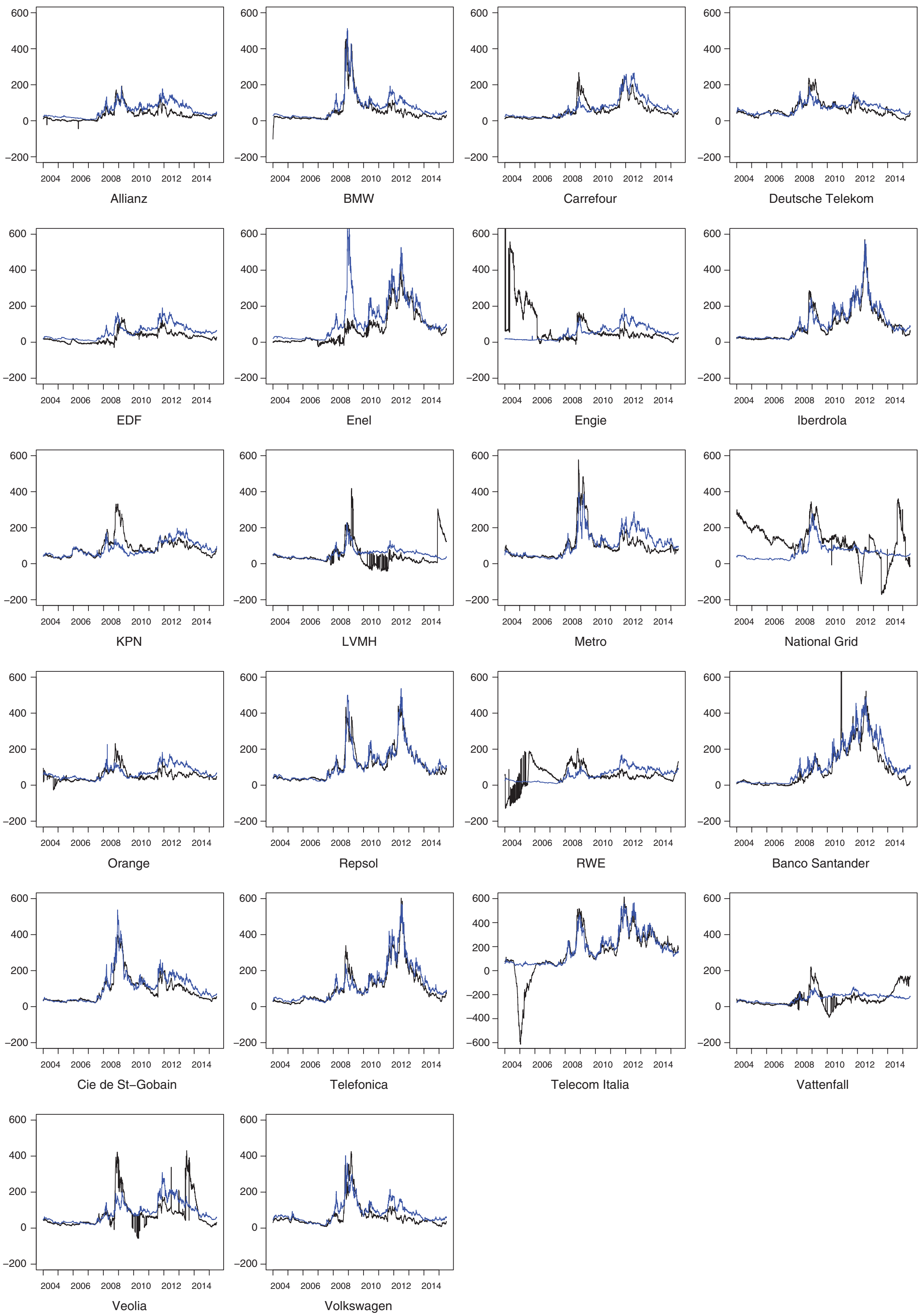

Figure 2: CDS and bond spread time series.
The figure shows the CDS (blue) and bond spread (black) time series for each reference entity.

same sign as the shock itself. As these shocks are generally associated with news arriving in the market, positive news are restricted to have a positive effect on the market. Furthermore, the contemporaneous effect of an idiosyncratic shock originating in market i ($e_{i,t}$) on the own market price ($p_{i,t}$) is greater than the contemporaneous effect on all other prices. Again, in terms of news, this means that the effect of news arriving is greatest in the home market. Based on these assumptions the information shares in Equation (5) are uniquely determined.[2]

3 Estimation strategy and data description

Estimation is performed in two steps. First, the VECM in Equation (1) is estimated by means of ordinary least squares. Using the residual time series, u_t, i.e. the composite innovations, estimates for the matrix of contemporaneous effects W and the second regime covariance matrix Ψ are obtained via Maximum-likelihood. Thereby, two strategies are pursued to obtain Maximum-likelihood estimates. In the first approach, the log-likelihood function is optimized directly. The alternative strategy is to use the Expectation-Maximization algorithm adapted to the Markov-switching framework (see Krolzig, 1997; Lanne, Lütkepohl, and Maciejowska 2010; Herwartz and Lütkepohl 2014). Standard errors for the estimated parameters are based on a parametric bootstrap (compare Davidson and MacKinnon 2000). Details concerning the estimation and bootstrap procedure are outlined in the Appendix.

Our data set includes daily CDS and bond time series of 22 iTraxx companies ranging from January 1, 2004 to June 30, 2015 (2987 observations). The sample includes those reference entities which are constituents of the iTraxx Europe index and for which sufficient CDS and bond data was available during the sample period, in order to construct CDS and bond spread times series. CDS premia data are obtained from Datastream. Following Blanco, Brennan, and Marsh (2005) and Doetz (2007), we use 5-year contracts as they are most liquid. The corresponding bond yield series are calculated from bonds with different maturities using linear interpolation, so that the constructed time series of bond yields matches the constant 5-years maturity of the CDS time series (for details see Dimpfl and Peter 2013). In order to avoid measurement errors due to various options in bonds, we only include bonds with a fixed rate that are not callable, puttable, or convertible. Finally we calculate bond spreads as the difference between the risky bond yields and the risk-free interest rate. We use 5-year swap rates as a proxy for the risk-free rate. Bond and swap time series are obtained from Bloomberg.

Table 1 provides details on our sample reference entities. There is one industrial company; two companies belong to the financial sector, two to the automobile sector; three companies are wholesalers; five companies belong to the telecommunications sector, and nine to the energy sector. Figure 2 depicts the CDS and the credit spread time series for each reference entity. As can be seen, all time series exhibit more or less two spikes during the sample period. The first is associated with the recent financial crisis (roughly between mid 2007 and the beginning of 2009). The second is most likely associated with the Greek debt crisis, which saw one of its peaks in 2012 when Fitch, Standard and Poor's, and Moody's downgraded Greek government bonds to the lowest level.

4 Empirical results

We now turn to estimating the information shares. First, we provide the results for the unique information shares based on the Markov-switching approach. We then compare these results to the standard approach of Hasbrouck (1995). In a second step we identify possible determinants of the estimated information shares.

2 The formal proof with Markov switching innovations works analogously to Grammig and Peter (2013).

Table 1: Reference entities.

Ticker	Company	Country	Sector
AZ	Allianz	Germany	Financials
BMW	BMW	Germany	Automobile
CAR	Carrefour	France	Consumers
DT	Deutsche Telekom	Germany	TMT
EDF	Electricite de France	France	Energy
EN	Enel	Italy	Energy
ENGI	Engie	France	Energy
IBE	Iberdrola	Spain	Energy
KPN	Koninklijke KPN	Netherlands	TMT
LVMH	LVMH	France	Consumers
MEO	Metro	Germany	Consumers
NGG	National Grid	UK	Energy
ORA	Orange	France	TMT
REP	Repsol	Spain	Energy
RWE	RWE	Germany	Energy
SGO	St Gobain	France	Industrials
STD	Banco Santander Central Hispano	Spain	Financials
TEF	Telefonica	Spain	TMT
TIT	Telecom Italia	Italy	TMT
VAT	Vattenfall	Sweden	Energy
VE	Veolia	France	Energy
VOW	VW	Germany	Automobile

The table presents the ticker symbols, company names, country, and industry sector of the sample reference entities.

4.1 Credit risk price discovery with two variance regimes

Table 2 holds the estimation results for the variances of the second regime, the transition probabilities and the unconditional probability for state 1 (the state 2 probability equals 1–p), as well as the CDS market information share (the bond market information share $IS_{\text{bond}}=100-IS_{\text{CDS}}$).

The estimated values for the diagonal elements of Ψ are larger than 1 for all companies but Enel. This shows that the second regime, which also allows for differing variances of the CDS and bond time series, is generally governed by a higher variance compared to the first regime. In case of Enel, the first regime which has unit variance by definition [see Equation (4)] is the regime with the higher variance.

In 13 cases the idiosyncratic innovations in the CDS market exhibit a relatively higher second regime variance compared to the bond market estimate $(\hat{\Psi}_1 > \hat{\Psi}_2)$. In the remaining eight cases the variance of the innovations in the bond market is higher. The estimated transition probabilities indicate that the regimes are not independent. p_{11} and p_{22} estimates are in general close to unity, which implies that conditional on being in a specific state in time t, the probability to remain in this state is high. These results correspond to previous findings by Alexander and Kaeck (2008). With the exception of Enel and Iberdrola, $P(s_t=1)\geq 0.55$ which indicates that during our sample period the two markets were predominantly governed by the low-volatility regime.

Turning to price discovery, the estimated information shares imply that overall the CDS market is the informationally dominant market. For 19 out of 22 reference entities the CDS information share exceeds 50%. For 11 reference entities the estimated information shares are above 90%, which indicates that price discovery takes place almost exclusively on the CDS market.

These results are in line with previous studies. However, they allow a much more accurate assessment of the leading market in price discovery. While, for instance, information shares' upper and lower bounds in Blanco, Brennan, and Marsh (2005) vary by up to 25 percentage points, our results deliver a unique measure. In addition, the Markov switching nature of our model allows us to examine the dynamic behavior of the information shares. By using the filtered and smoothed regime probabilities, we can generate a time series of information shares. Figure 3 depicts the development of CDS market information shares for each reference

Table 2: Estimation results.

Ticker	Ψ_B	Ψ_{CDS}	p_{11}	p_{12}	p_{21}	p_{22}	p	IS_{CDS}
AZ	11.61	41.04	0.94	0.07	0.06	0.93	0.55	69.87
	(0.87)	(2.38)	(0.22)	(0.00)	(0.00)	(0.22)	(0.00)	(22.08)
BMW	27.16	73.26	0.96	0.13	0.04	0.87	0.76	47.50
	(1.63)	(4.47)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(12.80)
CAR	14.65	38.43	0.95	0.11	0.05	0.89	0.68	67.97
	(0.79)	(2.14)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(23.52)
DT	10.08	21.76	0.96	0.12	0.04	0.88	0.75	64.32
	(0.62)	(1.31)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(21.61)
EDF	13.17	43.30	0.92	0.15	0.08	0.85	0.64	98.81
	(0.72)	(2.24)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(13.05)
EN	0.03	0.02	0.89	0.07	0.11	0.93	0.38	51.12
	(0.00)	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(26.22)
ENGI	373.22	24.27	0.96	0.40	0.04	0.60	0.91	99.89
	(34.97)	(2.09)	(0.00)	(0.00)	(0.00)	(0.03)	(0.00)	(13.63)
IBE	17.20	129.82	0.93	0.06	0.07	0.94	0.49	97.48
	(0.86)	(8.16)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(10.36)
KPN	11.72	17.67	0.94	0.18	0.06	0.82	0.73	99.74
	(0.65)	(1.01)	(0.01)	(0.00)	(0.00)	(0.02)	(0.00)	(15.78)
LVMH	236.97	26.82	0.96	0.15	0.04	0.85	0.78	98.94
	(17.87)	(1.61)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(10.29)
MEO	50.74	27.68	0.96	0.13	0.04	0.87	0.76	5.23
	(3.04)	(1.60)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(22.91)
NGG	31.12	34.95	0.94	0.19	0.06	0.81	0.75	95.93
	(1.88)	(2.24)	(0.01)	(0.00)	(0.00)	(0.02)	(0.00)	(28.99)
ORA	20.53	44.26	0.95	0.16	0.05	0.84	0.78	96.33
	(1.27)	(2.84)	(0.00)	(0.00)	(0.00)	(0.02)	(0.00)	(16.12)
REP	37.03	29.52	0.95	0.13	0.05	0.87	0.70	1.48
	(2.19)	(1.71)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(11.38)
RWE	206.41	8.82	0.97	0.19	0.03	0.81	0.85	88.42
	(15.58)	(0.59)	(0.00)	(0.00)	(0.00)	(0.02)	(0.00)	(9.25)
STD	62.39	30.26	0.95	0.10	0.05	0.90	0.66	83.67
	(3.39)	(1.61)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(5.49)
SGO	16.41	49.14	0.96	0.07	0.04	0.93	0.62	58.84
	(0.92)	(2.85)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(17.46)
TEF	14.50	52.99	0.94	0.08	0.06	0.92	0.59	99.31
	(0.78)	(3.02)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(11.45)
TIT	17.25	25.31	0.91	0.13	0.09	0.87	0.60	97.09
	(0.91)	(1.31)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(30.09)
VAT	69.31	15.79	0.95	0.18	0.05	0.82	0.79	97.09
	(4.63)	(0.99)	(0.00)	(0.00)	(0.00)	(0.02)	(0.00)	(15.83)
VE	231.81	21.70	0.95	0.12	0.05	0.88	0.69	98.02
	(16.56)	(1.24)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	(8.66)
VW	34.39	42.48	0.96	0.11	0.04	0.89	0.75	65.51
	(2.02)	(2.76)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	(25.07)

The table shows the estimation results for the second regime's variances of the idiosyncratic innovations of the bond and CDS markets (Ψ_B, Ψ_{CDS}), the transition probabilities (p_{ij}), and the unconditional probability of the first regime (p). The last column contains the estimated CDS market information shares (IS_{CDS}). Standard errors derived by a parametric bootstrap are in parentheses.

entity during the sample period. The graphs show that for several reference entities, such as BMW, Carrefour, or Iberdrola, the CDS information share in the second state is considerably higher than during the first regime. For these reference entities the regime switches to the high-volatility state around the onset of the financial crisis and remains predominantly in the high-volatility regime until the end of 2011. The importance of the CDS market compared to the bond market obviously increases during periods associated with a higher volatility.

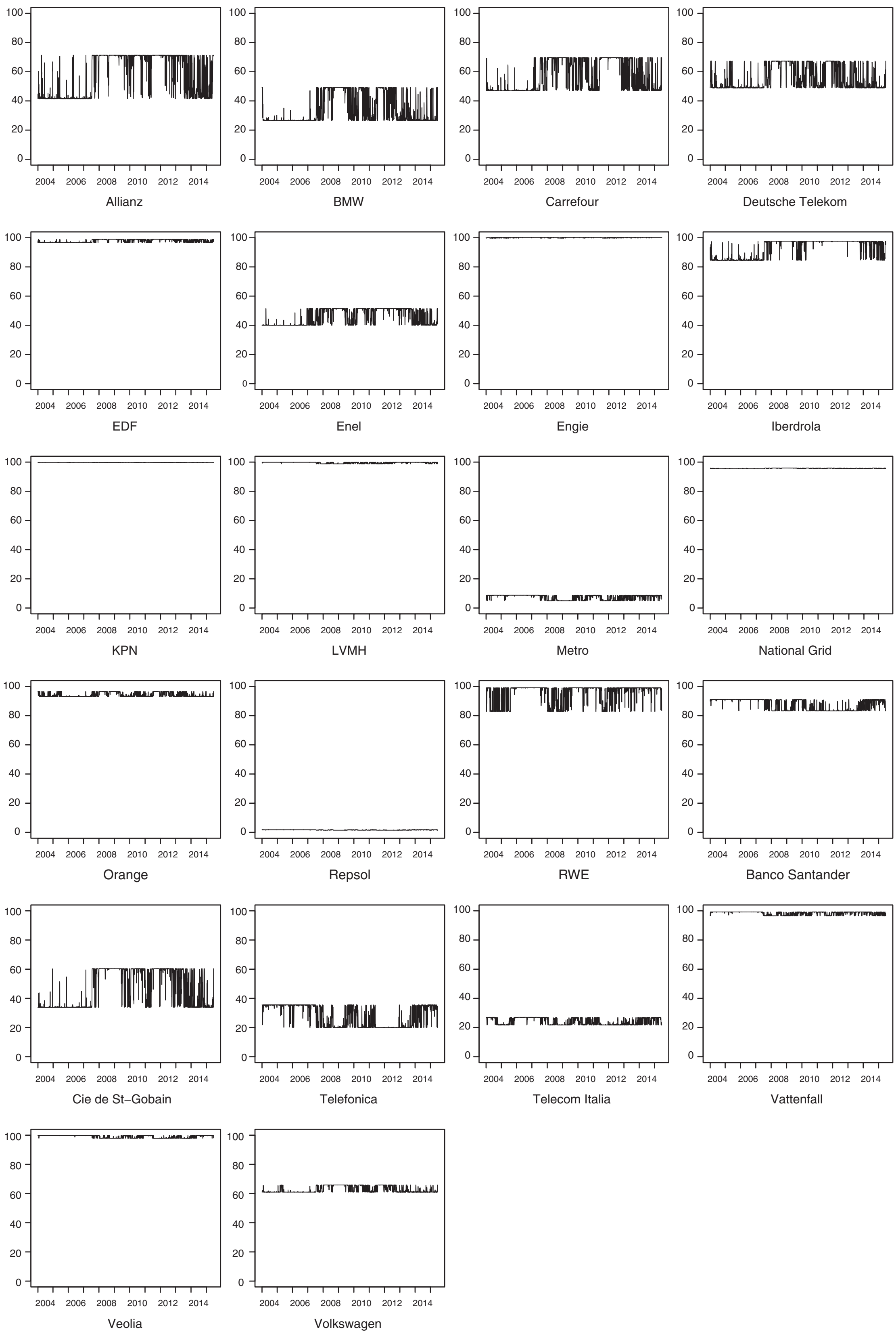

Figure 3: Filtered CDS market information share time series.
The figure presents the time series of the filtered CDS market information shares for each reference entity. The information share time series were calculated by using the smoothed unconditional regime probabilities.

We further examine this issue by calculating regime-specific information shares. Note that the following is not generally in line with the Hasbrouck (1995) methodology and the underlying microstructure model. However, in this particular application, it is sensible to assume that the pre-crisis period is dominated by the low-volatility regime and the crisis and post-crisis periods by the high-volatility regime. Therefore calculating regime-dependent information shares can yield additional insights.

We derive the regime-dependent information share for the the first regime by setting p=1 in Equation (5). Analogously, setting p=0 in Equation (5) yields the information share for the second regime. Results for the CDS market information shares are displayed in Table 3. As before, the information share of the bond market is given by $100-IS_{\text{CDS}}$.

The results show that the importance of the CDS market is larger during the high-volatility regime compared to the low-volatility regime for most reference entities. Exceptions are Enel, Repsol, Vattenfall, and Veolia, for which the CDS market information share is slightly lower in the second regime. For Engie, KPN, National Grid, or LVMH we find information shares of the CDS market close to unity, which implies that the CDS market is by far the informationally dominant market in both regimes.

A possible explanation for these findings is increased insider trading in the CDS market during the crisis period as reported by Coro, Dufour, and Varotto (2013). They explore the credit and liquidity determinants of CDS price movements and investigate how their role changed as a result of the 2007–2009 financial crisis. They find that informed trading has a more prominent impact on CDS prices during the crisis period. However, Coro, Dufour, and Varotto (2013) examine CDS prices only. In order to determine whether informed traders actually prefer trading on the CDS market rather than trading on the bond market during crisis periods, the changes of the extent of informed trading on both markets would have to be analyzed. Nevertheless, by comparing overall CDS information shares to the regime-dependent information shares, it becomes obvious that the results are driven by the high-volatility regime. Our analysis therefore emphasizes the importance of the CDS market for price discovery during crisis periods.

Table 3: Regime specific information shares.

Ticker	IS_{CDS}	$IS_{\text{CDS}}^{\text{Regime 1}}$	$IS_{\text{CDS}}^{\text{Regime 2}}$
AZ	69.87	41.30	71.32
BMW	47.50	26.41	49.20
CAR	67.97	46.73	69.71
DT	64.32	48.79	67.28
EDF	98.81	96.51	98.91
EN	51.12	51.51	39.89
ENGI	99.89	99.99	99.85
IBE	97.48	84.32	97.60
KPN	99.74	99.63	99.76
LVMH	98.94	99.86	98.81
MEO	5.23	5.00	8.79
NGG	95.93	95.49	95.97
ORA	96.33	92.97	96.61
REP	1.48	1.82	1.46
RWE	88.42	99.12	82.79
SGO	58.84	33.71	60.35
STD	83.67	83.24	91.10
TEF	99.31	97.70	99.36
TIT	97.09	95.90	97.17
VAT	97.09	99.21	96.61
VE	98.02	99.79	97.84
VW	65.51	60.97	65.86

The table shows the CDS market information share (column 2) together with regime-specific information shares (columns 3 and 4). Regime 1 corresponds to the low-volatility period and regime 2 to the high-volatility period.

4.2 Comparison with the standard Hasbrouck methodology

As a robustness check for our methodology we also compute Hasbrouck's (1995) information shares using Cholesky decomposition [as outlined in Equation (3)] instead of the variance regimes to identify the contribution of each market to price discovery. Table 4 presents upper and lower bounds of the Hasbrouck information share HIS along with their average and, for comparison, the information share reported in Table 2.

For 19 out of the 22 reference entities, the information shares based on the regime-switching identification strategy are either close to or higher than the average of the Hasbrouck information shares. Exceptions are Metro, Repsol, and St Gobain. For these three companies, our information share is noticeably lower than the Hasbrouck lower bound. Still, the conclusions drawn are the same: in the case of Metro and Repsol, the bond market is found to be the dominant market based on both methodologies, while in case of St Gobain it is the CDS market that leads price discovery. We are therefore confident that our identification strategy delivers accurate, unique information shares that are in line with the methodological market microstructure framework of Hasbrouck (1995).

4.3 Information share determinants

In order to shed light on the question about possible determinants of the information shares, we conduct a cross-sectional analysis. To this end we regress the CDS market information shares on company-specific characteristics. As information shares lie between zero and one, we apply the logit transformation $\ln(IS_{CDS}/(1-IS_{CDS})$ to construct the dependent variable.

As the first explanatory variable, we use relative liquidity of the CDS and bond market, as liquidity is often identified as the most important determinant of the information shares (compare Yan and Zivot 2010).

Table 4: Information share comparison.

Ticker	IS_{CDS}	HIS_{low}	HIS_{up}	HIS_{avg}
AZ	69.87	67.03	69.65	68.34
BMW	47.50	18.05	34.45	26.25
CAR	67.97	59.14	62.88	61.01
DT	64.32	66.68	67.45	67.07
EDF	98.81	98.01	98.26	98.14
EN	51.12	35.15	44.75	39.95
ENGI	99.89	99.97	99.99	99.98
IBE	97.48	76.90	96.65	86.77
KPN	99.74	99.51	99.71	99.61
LVMH	98.94	98.17	98.75	98.46
MEO	5.23	18.09	38.03	28.06
NGG	95.93	79.95	81.22	80.59
ORA	96.33	91.40	94.63	93.01
REP	1.48	29.72	54.17	41.94
RWE	88.42	88.94	89.46	89.20
SGO	58.84	88.22	94.86	91.54
STD	83.67	58.15	61.31	59.73
TEF	99.31	72.12	99.69	85.90
TIT	97.09	78.51	98.94	88.73
VAT	97.09	97.36	98.60	97.98
VE	98.02	98.32	99.53	98.93
VW	65.51	52.13	62.40	57.26

The table presents the CDS market information shares based on the Markov switching approach (IS_{CDS}) along with the standard Hasbrouck methodology, where HIS_{low} (HIS_{up}) is the lower (upper) bound of the information share and HIS_{avg} is the average.

We construct a proxy measure as the ratio between CDS and bond market bid-ask spreads.[3] We also include the company's market capitalization and the debt/equity ratio as a measure of a company's financial leverage. These data are obtained from Compustat. Lastly, we add dummy variables for the financial sector (FIN), telecommunications (TMT), or energy (Energy).

Table 5 holds the resulting parameter estimates and the corresponding heteroskedasticity robust standard errors. Each column corresponds to one regression and we successively add more explanatory variables. In the first regression we only include relative liquidity. The negative estimate implies that decreasing liquidity in the CDS relative to the bond market (i.e. an increasing CDS market spread and/or decreasing bond market spread) is associated with a lower CDS market information share. This conclusion is in line with the positive link between liquidity and contributions to price discovery derived in previous studies (see Chen, Choi, and Hong 2013; Kehrle and Peter 2013).

We then add the debt/equity ratio and market capitalization. The measure for relative liquidity is again negative, but this time it is only significant on the 10% significance level. The debt/equity ratio and market capitalization are not statistically significant. They still seem to improve the model as the adjusted R^2 increases to 0.47 (as compared to 0.39 in the first estimation). The coefficient on the debt/equity ratio is found to be positive, which would imply that higher leverage increases the CDS market information share relative to the bond market share. Similarly, a higher market capitalization would imply a higher CDS market information share.

Lastly, including the industry dummy variables leads to a lower adjusted R^2, which suggests that the previous two models are superior and that the attribution to an industry sector does not explain the magnitude of the information shares. The fact that statistical significance is very weak in general is most likely due to the fact that we only have 22 observations. Still, liquidity remains a significant determinant of the information shares in this model as well.

We also conduct the analysis for the regime-dependent information shares using all firm characteristics besides the industry dummy variables as explanatory variables. Table 6 presents the results along with the result for the unconditional information share.

Table 5: Information share determinants.

Variable	IS_{CDS}	IS_{CDS}	IS_{CDS}
Intercept	6.30***	4.03**	4.56
	(1.31)	(1.68)	(2.90)
Liquidity	−0.27***	−0.20*	−0.23*
	(0.09)	(0.10)	(0.13)
Debt/equity		0.94	0.62
		(1.01)	(0.86)
Market cap.		0.01	0.01
		(0.01)	(0.01)
TMT			01.63
			(1.62)
FIN			−1.32
			(1.47)
Energy			−0.62
			(1.83)
Adjusted R^2	0.39	0.47	0.37

The table presents the results for the regressions of the logit-transformed CDS market information shares on firm characteristics. The base category in the third model includes all remaining industries represented in the sample (industrials, automobiles, consumers). Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level of significance, respectively.

3 Bid-ask spreads in the CDS market are calculated from daily bid and ask prices. Bond market bid-ask spreads are calculated as the average over the daily bid-ask spreads of all active bond time series using daily bid and ask prices.

Table 6: Regime specific information share determinants.

Variable	IS_{CDS}	$IS_{CDS}^{Regime 1}$	$IS_{CDS}^{Regime 2}$
Intercept	4.03**	4.96**	3.58*
	(1.68)	(2.01)	(1.72)
Liquidity	−0.20*	−0.25**	−0.18*
	(0.10)	(0.10)	(0.10)
Debt/equity	0.94	0.80	0.76
	(1.01)	(1.15)	(0.71)
Market cap.	0.01	0.01	0.01
	(0.01)	(0.01)	(0.01)
Adjusted R^2	0.47	0.18	0.22

The table presents the results for the regressions of the logit-transformed CDS market information shares on firm characteristics. IS_{CDS} is the unconditional information share, while $IS_{CDS}^{Regime 1}$ and $IS_{CDS}^{Regime 2}$ are the information shares for the regimes 1 and 2. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% level of significance, respectively.

The results show that relative liquidity is an important factor regardless of the volatility regime. The estimate is negative and statistically significant in all models. The debt/equity ratio and market capitalization are again not found to have any explanatory power. This suggests that even though the information shares differ across regimes, what determines them does not.

5 Conclusion

We propose a Markov switching framework to model price discovery between the CDS and the corporate bond market. The major advantage of our approach lies in delivering a unique measure for the markets' contribution to the price discovery process as opposed to the Hasbrouck (1995) measure, which yields a range. At the same time, the Markov property allows for a regime-switching behavior of information shocks in the two markets.

Analyzing a sample of 22 iTraxx companies, we determine the CDS market as the informationally dominant market. The CDS market information share even increases in periods marked by high volatility. We also examine potential determinants for the information share measures. The results support previously reported evidence of a positive link between liquidity and information shares. Firm-specific characteristics such as market capitalization, leverage, or affiliation to a specific industry are not found to be statistically significant. All in all, our results emphasize the role of the CDS market for the price discovery process of credit risk, in particular during highly volatile, i.e. crisis periods.

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Appendix

A Construction of the likelihood function

Denote the vector of price series by y_t. Conditional on the unobserved state variable s_t, the density function of y_t is known. As the innovations are assumed to be conditionally normally distributed, the conditional density of y_t is given by

$$f(y_t|s_t=j,\ Y_{t-1})=(2\pi)^{-\frac{n}{2}}\det(\Sigma_j)^{-\frac{1}{2}}\exp\left(-\frac{1}{2}u_t\Sigma_j^{-1}u_t\right) \tag{A.1}$$

for j=1, 2 and $Y_{t-1}=\{y_1, \ldots, y_{t-1}\}$. Multiplying the conditional density in Equation (A.1) with the marginal density of s_t yields the joint density of y_t and s_t

$$f(y_t,\ s_t=j|Y_{t-1})=\Pr(s_t=j|Y_{t-1})f(y_t|s_t=j,\ Y_{t-1}) \tag{A.2}$$

for j=1, 2. Summing over all regimes allows to integrate out s_t, which results in the unconditional density of y_t:

$$f(y_t|Y_{t-1})=\sum_{j=1}^{2}f(y_t,\ s_t=j|Y_{t-1}). \tag{A.3}$$

This density corresponds to the average of the conditional densities weighted by the probability of the particular regime. To compute the weighting factors, we have to account for the Markov structure of s_t. In time t, the probability of regime j can be calculated as $\Pr(s_t=j|Y_{t-1})=\Pr(s_t=j|s_{t-1}=i)\times\Pr(s_{t-1}=j|Y_{t-1})$, where the probabilities in the first term on the right hand side are summarized in the transition matrix $P=\begin{bmatrix} p_{11} & 1-p_{22} \\ 1-p_{11} & p_{22} \end{bmatrix}$ with $p_{ij}=\Pr(s_t=j|s_{t-1}=i)\ \forall i, j\in 1, 2$.

Using Bayes' theorem, the probability $\Pr(s_t=j|Y_t)$ can be filtered out, which together with the transition matrix P delivers $\Pr(s_{t+1}=j|Y_t)$. Iterating these steps until the last observation generates $f(y_t|Y_{t-1})$ for each point in time. Finally, the log-likelihood function is given by

$$logL_T=\sum_{t=1}^{T}log\, f(y_t|Y_{t-1}). \tag{A.4}$$

It is maximised with respect to W, Ψ, p_{11}, and p_{22}.

The main difficulty of this optimization approach concerns the search of appropriate initial values. Due to the complexity of the log-likelihood function, finding the global peak proves to be challenging. In order to verify the maximization results and to find a more reliable method in terms of starting values, we recommend to apply the EM-algorithm.

B The EM-algorithm

In this study, the approach described by Herwartz and Lütkepohl (2014) is implemented. It represents the EM-algorithm enhanced by a Baum-Lindgren-Hamilton-Kim (BLHK) filter, which is explained by Krolzig (1997). Contrary to the approach of Herwartz and Lütkepohl (2014), we do not compute the residual series within the EM-algorithm, but estimate the VECM model in Equation (1) and subsequently use the VECM residuals in the optimization of the log-likelihood.

Let P denote the transition matrix, ι is a 2×1 vector of ones, and

$$\xi_{t|s}=\begin{bmatrix} \Pr(s_t=1|Y_s) \\ \Pr(s_t=2|Y_s) \end{bmatrix},$$

$\odot$ stands for element-wise multiplication,
$\oslash$ stands for element-wise division and
$\otimes$ for the Kronecker product.

1. Initialization of the starting values:

   $P^{(0)}$, $\xi_{0|0}$, W^0 and $\Psi^{(0)}$.

2. Expectation step:
   The expectation step of the EM-Algorithm begins with the filtering or a "*forward recursion*"

$$\xi_{t|t} = \frac{\eta_t \odot P\xi_{t-1|t-1}}{\iota'(\eta_t \odot P\xi_{t-1|t-1})} \text{ for } t=1, \ldots, T,$$

   where

$$\eta_t = \begin{bmatrix} f(y_t \,|\, s_t=1, Y_{t-1}) \\ f(y_t \,|\, s_t=2, Y_{t-1}) \end{bmatrix},$$

   with

$$f(y_t \,|\, s_t=m, Y_{t-1}) = (2\pi)^{-\frac{K}{2}} \det(\Sigma_m)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2} u_t' \Sigma_m^{-1} u_t\right\} \text{ for } m=1, 2.$$

   Then, the filtered probabilities are smoothed by means of "*backward recursion*":

$$\xi_{t|T} = (P'(\xi_{t+1|T} \oslash P\xi_{t|t})) \odot \xi_{t|t} \quad \text{for } t=T-1, \ldots, 0.$$

3. Maximization step:
   Along the maximization step, the transition matrix is estimated with the Hidden Markov Chain formula (Krolzig 1997)

$$vec(\hat{P}') = \left(\sum_{t=1}^{T-1} \xi_{t|T}^{(2)}\right) \oslash \left(\iota \otimes \sum_{t=1}^{T} \xi_{t|T}\right)$$

   where

$$\xi_{t|T}^{(2)} = vec(P') \odot [(\xi_{t+1|T} \oslash P\xi_{t|t}) \otimes \xi_{t|t}] \quad \text{for } t=0, \ldots, T-1.$$

   Finally, the values of W and Ψ, which maximize the expected log-likelihood function, have to be found. The expected log-likelihood is given by

$$\log L_{EM} = -\frac{1}{2} \sum_{t=1}^{T} \sum_{m=1}^{2} \xi_{mt|T} [2\log(2\pi) + \log(|\Sigma_m|) + u_t' \Sigma_m^{-1} u_t].$$

   Writing out the regimes yields

$$= -\frac{1}{2}\Bigg[2\log(2\pi)\sum_{t=1}^{T}\xi_{1t|T} + \sum \xi_{1t|T}\log(|WW'|) + tr\left((WW')^{-1}\sum_{t=1}^{T}\xi_{1t|T}\hat{u}_t'\hat{u}_t\right) + 2\log(2\pi)\sum_{t=1}^{T}\xi_{2t|T}$$
$$+ \sum \xi_{2t|T}\log(|W\Psi W'|) + tr\left((W\Psi W')^{-1}\sum_{t=1}^{T}\xi_{2t|T}\hat{u}_t'\hat{u}_t\right)\Bigg].$$

The likelihood function is then maximized subject to constraints on the W and Ψ matrices discussed in Section 2. In addition, a lower bound of 0.001 is imposed on the determinant of each covariance matrix. The estimates become the initial values for the next iteration:

$$\hat{P} = P^{(0)},$$
$$\xi_{0|T} = \xi_{0|0},$$
$$\hat{W} = W^{0}, \text{ and}$$
$$\hat{\Psi} = \Psi^{(0)}.$$

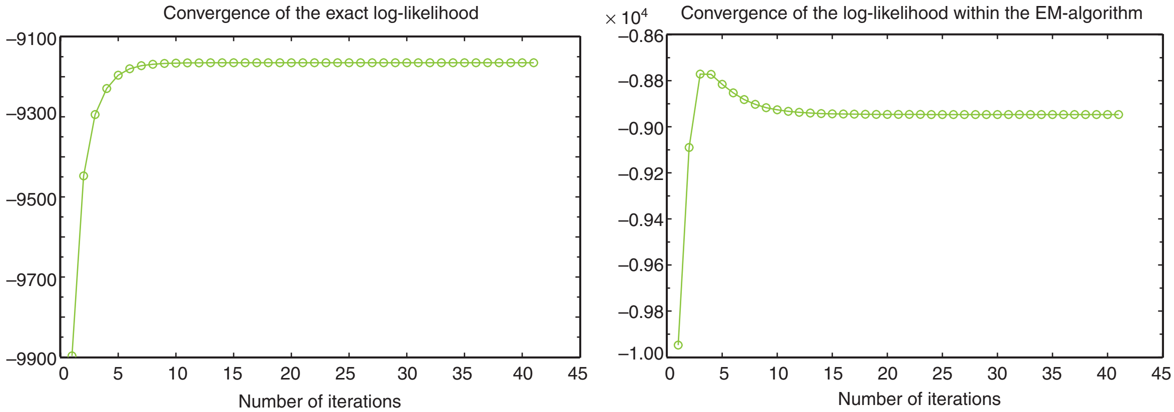

Figure B.1: Convergence of the log-likelihood function.
The graphic illustrates the convergence of the log-likelihood function for the estimation of Allianz.

Steps 2 and 3 are iterated until convergence is achieved, e.g. until

$$\sum_{i=1}^{2}\sum_{j=1}^{2}|\hat{W}_{ij}-W_{ij}^{(0)}|<0.00001.$$

Unfortunately, using numerical optimization does not guarantee monotonic convergence of the expected log-likelihood Schaefer (1997). This is the case in our study as shown by Figure B.1. It shows that using numerical optimization leads to non-monotonic convergence in our empirical application. Consequently we verify our results by applying both approaches, direct maximization of the loglikelihood function an the EM-algorithm outlined above.

Standard errors of the estimates are obtained by means of a parametric bootstrap. This bootstrap is a slightly adapted version of the procedure described in Grammig and Peter (2013). It proves to be convenient, bearing in mind the two-step estimation structure.

Estimation is done in Gauss and all program codes are available upon request.

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